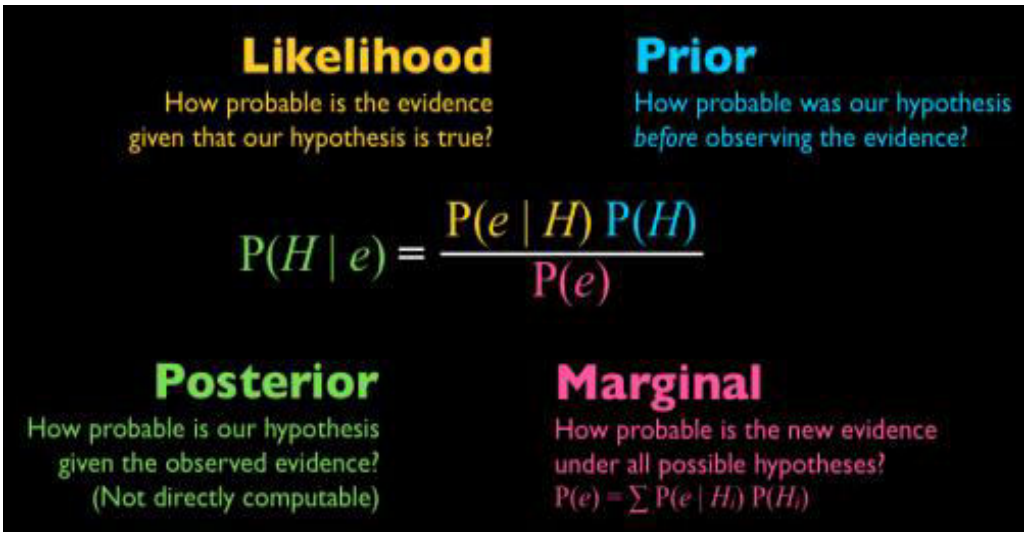
Naïve Bayes Classification

1. Introduction

What is Naïve Bayes?

Naïve Bayes is a probabilistic classification algorithm based on Bayes' Theorem. It assumes independence between features, which makes it computationally efficient. It is widely used in applications such as spam detection, sentiment analysis, and medical diagnosis.

2. Understanding Bayes' Theorem



Formula:

$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$ Where:

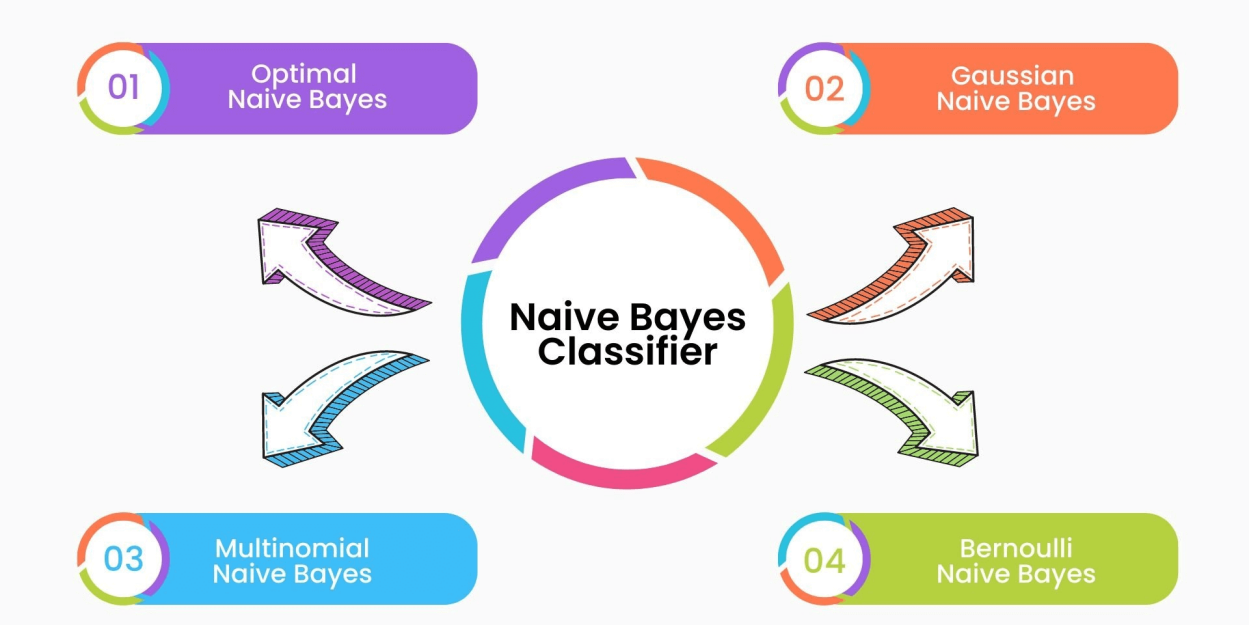
- $P(A|B)$: Posterior probability (probability of event A given event B)
- $P(B|A)$: Likelihood (probability of event B given event A)
- $P(A)$: Prior probability of event A
- $P(B)$: Probability of event B

Example Calculation:

Suppose 2% of the population has a disease ($P(A)$=0.02), and a test detects it 95% of the time ($P(B|A)$=0.95), but also has a 5% false positive rate ($P(B)$=0.05). In that case, actually having the disease given a positive test.

3. Types of Naïve Bayes Classifiers

1. **Bernoulli Naïve Bayes** - Used for binary feature data (e.g., spam filtering)
2. **Multinomial Naïve Bayes** - Used for text classification (e.g., sentiment analysis)
3. **Gaussian Naïve Bayes** - Used for continuous data (e.g., medical diagnosis)



4. **Implementation in Scikit-Learn**

```python
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import BernoulliNB, MultinomialNB, GaussianNB
from sklearn.metrics import accuracy_score

# Sample Dataset
X = np.array([[1, 0, 1], [0, 1, 0], [1, 1, 1], [0, 0, 0]])  # Features
y = np.array([1, 0, 1, 0])  # Labels

# Splitting data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Bernoulli Naïve Bayes
bnb = BernoulliNB()
bnb.fit(X_train, y_train)
y_pred = bnb.predict(X_test)
print("Bernoulli Naïve Bayes Accuracy:", accuracy_score(y_test, y_pred))

# Multinomial Naïve Bayes
mnb = MultinomialNB()
mnb.fit(X_train, y_train)
y_pred = mnb.predict(X_test)
print("Multinomial Naïve Bayes Accuracy:", accuracy_score(y_test, y_pred))

# Gaussian Naïve Bayes
gnb = GaussianNB()
gnb.fit(X_train, y_train)
y_pred = gnb.predict(X_test)
print("Gaussian Naïve Bayes Accuracy:", accuracy_score(y_test, y_pred))
```

5. Applications of Naïve Bayes

- **Spam Filtering**: Classifies emails as spam or not
- **Sentiment Analysis**: Determines sentiment in text
- **Disease Prediction**: Predicts likelihood of diseases
- **Recommender Systems**: Used in movie and product recommendations

6. Advantages & Disadvantages

Advantages:

✅ Fast and efficient for large datasets ✅ Works well with small datasets ✅ Handles missing data well

Disadvantages:

❌ Assumes feature independence, which is unrealistic in many cases ❌ Performs poorly with highly correlated features ❌ Sensitive to small datasets and class imbalances

7. Conclusion

Naïve Bayes is a powerful classification algorithm based on probability theory. It is best suited for text classification and applications where feature independence is a reasonable assumption. Choosing the right variant (Bernoulli, Multinomial, Gaussian) depends on the nature of the dataset.

8.Interview Questions & Answers:

1. **What is Naïve Bayes? Explain its assumptions.**
    - Naïve Bayes is a probabilistic classifier based on Bayes' Theorem.
    - Assumes feature independence.

2. **How does the Naïve Bayes classifier handle missing values?**
    - It can handle missing values by ignoring them during probability computation.

3. **What are the types of Naïve Bayes algorithms?**
    - Bernoulli, Multinomial, and Gaussian.

4. **When would you use Gaussian Naïve Bayes?**
    - When features are continuous and normally distributed.

5. **What are the advantages of Naïve Bayes?**
    - Fast computation, works well with small data, robust to noise.

6. **What are the limitations of Naïve Bayes?**
    - Assumes feature independence, struggles with highly correlated data.

7. **How does Naïve Bayes compare with other classifiers like SVM or Decision Trees?**
    - Faster but less accurate on complex datasets.

8. **How do you improve the accuracy of a Naïve Bayes model?**
    - Feature selection, using ensemble methods.

9. **What are some real-world applications of Naïve Bayes?**
    - Spam filtering, sentiment analysis, medical diagnosis.

10. **How can you implement Naïve Bayes in Scikit-learn?**
    - Using BernoulliNB, MultinomialNB, GaussianNB classes.

9. Kaggle & GitHub Projects for Learning Naïve Bayes

Kaggle Projects:

1. https://www.kaggle.com/datasets/jackksoncsie/spam-email-dataset
2. https://www.kaggle.com/code/lakshmi25npathi/sentiment-analysis-of-imdb-movie-reviews
3. https://www.kaggle.com/code/ismailsefa/heart-disease-predic-machine-learning-naive-bayes

GitHub Projects:

1. **Text Classification using Naïve Bayes**
2. **Spam Detection using Naïve Bayes**
3. **Naïve Bayes in Scikit-learn**